Basel III requires banks to hold Tier 1 Capital of at least 6% of risk-weighted assets, with at least 4.5 percentage points held as CET1 Capital.

As a simplified example, if a bank wanted to extend additional loans to customers while its CET1 Capital was 4.5% of RWAs, the bank would need to fund a portion of the additional loans with common equity or retained earnings, so that its CET1 Capital ratio as a percentage of RWAs didn't drop below 4.5%.

As an alternative to raising additional CET1 capital to fund a portion of these additional loans, the bank could dispose of some existing risk-weighted assets on its balance sheet, making room for the new loans. In this case, the relevant cost of funds should reflect the shadow cost of the capital constraint. For example, if the bank disposes of assets with a return on capital of 7% in order to make room for new loans, then 7% is the shadow cost of the capital constraint that should be included in the bank's decision-making process. Even if the bank could fund the new loans at a marginal cost of, say, 4%, it must also consider the 7% return on capital it *won't* be earning if it disposes of existing assets in order to make room for the new loans.

Leverage Requirements

Regulators have concerns that their capital ratio requirements may not be sufficient to provide adequate protection against the risk of bank crises, so they introduced two additional, related measures: the *Tier 1 Leverage Ratio*, and the *Supplementary Leverage Ratio*.[4]

The Tier 1 Leverage Ratio is defined as the amount of Tier 1 Capital as a percentage of total assets, while the Supplementary Leverage Ratio is defined as the ratio of Tier 1 Capital to the *Exposure Measure*, which is the sum of on-balance sheet exposures, derivative exposures, and security financing exposures (e.g. repo transactions).

Different banks are subject to different requirements for these leverage ratios. The Basel III accords set the minimum Tier 1 Leverage ratio at 3%, but the specific regulations in many jurisdictions are both more complicated and more restrictive. For example, *Globally Systemically Important Banks* (G-SIBs) in the US are required to maintain a Tier 1 Leverage ratio of 6.71% and a Supplementary Leverage ratio of 5.64% currently.[5]

[4]For more on this, see "Basel IV: The Leverage ratio – Background and timeline of development" on www.pwc.co.uk.

[5]These leverage requirements are subject to change at the discretion of regulators.

A bank may find that one of the relevant leverage ratios is binding even if its capital ratios are not. In that case, the bank would be subject to considerations similar to those discussed in the context of capital ratios. In particular, a bank could find that its shadow cost of capital for a new investment is considerably greater than the marginal cost of funds that could be obtained via deposits or interbank loans, owing to the fact that an attractive asset with a higher return on capital may have to be liquidated in order to make room for the new investment.

Funding and Liquidity Requirements

In addition to the risk of insolvency, banking regulators are concerned with the risk of banks experiencing illiquidity, the inability to obtain cash when needed. To address the ability of banks to raise cash, regulators introduced the Net Stable Funding Requirement. And to address the ability of banks to convert assets to cash, regulators introduced the Liquidity Coverage Ratio.

Net Stable Funding Ratio

The Net Stable Funding Ratio (NSFR) is defined as the amount of stable funding available over a one-year horizon divided by the amount of stable funding required over a one-year horizon. The definitions of these two quantities are fairly involved. Quoting from the Federal Reserve Board of Governors' *Final rule to implement a net stable funding ratio requirement for large banking organizations*:

> [A] banking organization's available stable funding amount, the numerator of the NSFR, would measure the stability of its regulatory capital elements and liabilities. Regulatory capital elements and liabilities would each be assigned an available stable funding factor, which represents the extent to which the capital element or liability is considered available for use by the banking organization over a one-year time horizon. The available stable funding factors are scaled from zero (least stable) to 100 percent (most stable) and were determined by taking into account the tenor of the funding, type of funding, and type of counterparty.

The denominator is also addressed in the same document: "The required stable funding amount is the sum of the required stable funding amount for non-derivative assets and commitments and the required stable funding amount for derivative transactions."

The actual implementation of the NSFR requirement is also quite involved, with various assets and funding sources weighted by different degrees. But the key point for relative value analysts is that banks may find themselves unable to pursue various opportunities, including relative value trades, if pursuit of the opportunity were to increase the required stable funding requirement to the point that the NSFR declined to less than 100%. In such a case, the shadow cost of the NSFR constraint would need to be included in the relative value analysis.

Liquidity Coverage Ratio

The liquidity coverage ratio is defined as the amount of High Quality Liquid Assets on a bank's balance sheet divided by the net cash outflows over the next 30 calendar days. Banks must maintain this ratio at 100% or greater.

A High Quality Liquid Asset (HQLA) is one that can be quickly sold or pledged as collateral in a loan without an adverse effect on its price. The characteristics required for an asset to be considered an HQLA are not just those related to the asset per se but also to the terms governing its use on a bank's balance sheet. For example, according to the Bank for International Settlements:

> Banks must not include in the stock of HQLA any assets, or liquidity generated from assets, they have received under right of rehypothecation, if the beneficial owner has the contractual right to withdraw those assets during the 30-day stress period.

So in determining whether a particular asset can be counted as an HQLA, the bank must look to the rights of the counterparty that provided the asset as part of a collateral agreement.

Market Risk Regulations

Following the financial crisis of 2008–2009, banking regulators coordinated on a set of reforms that they referred to as the "Fundamental Review of the Trading Book" (FRTB), intended to lessen the risks that banks faced in their trading books, which need to be marked-to-market (as opposed to their banking books, which do not need to be marked-to-market).

The FRTB is a complicated process with a myriad of new procedures that banks must follow. Of these, two are particularly important from our perspective as relative value analysts.

First, the FRTB modifies the way banks need to calculate and report market risks in their trading books. For example, banks must shift from a focus on Value at Risk to a focus on Expected Shortfall. And, second, they must rely more on standard models and less on internal models than they used to. Taken together, the effect of these reforms is that most banks will be expected to hold greater capital in the future, increasing their overall costs of doing business.

From our perspective, these reforms appear likely to affect the marginal costs relevant to analyzing a relative value opportunity. For example, a simple spread trade that appeared attractive previously, based on unconstrained financing costs, may no longer appear attractive, once the costs of heightened capital requirements are included.

Example: JP Morgan and the Repo Spike of September 2019

On September 17, 2019, overnight repo rates in the US spiked from roughly 2.4% to intraday levels as high as 10%. In the past, we would have expected large banks to lend cash in repo markets in order to take advantage of such an unusually attractive opportunity. But according to JP Morgan CEO, Jamie Dimon, new bank regulations constrained his bank from acting. According to Bloomberg:

> *When rates on repurchase agreements spiked to around 10 per cent a month ago — roughly four times more than what JPMorgan earns at the Fed — the bank could've profited by shifting the money into repo.*
>
> *It didn't. The bank, Dimon told analysts following JPMorgan's third-quarter earnings release, needed to keep that money put so it could fulfill its liquidity requirements mandated by regulators.*
>
> *"We could not redeploy it into the repo market. We'd have been happy to do it," Dimon said Tuesday. "It's up to the regulators to decide if they want to recalibrate the kind of liquidity they expect us to keep in that account."*
>
> *This couldn't be more different than year-end 2018. Back then, JPMorgan deployed excess cash to the repo market when rates surged above six per cent as other lenders retreated as a way of tidying up their balance sheets for regulatory purposes. JPMorgan drew down its deposits with the Fed and increased its allocations to the repo market by more than US$100 billion, according to its fourth-quarter earnings statement.*
>
> *"Last year, we had more cash than needed for regulatory requirements," Dimon said. So shifting into repo "obviously made sense, you make more money," the CEO added.*[6]

In the end, the repo spike experienced that day didn't result in serious disruption to the banking system, in part because the Federal Reserve acted

[6]BNN Bloomberg, "JPMorgan felt barred from calming repo market by regulations: Dimon."

to provide liquidity in the repo market for the rest of the week. But the large distortions in the money markets certainly had a significant effect on the pricing of relative value trades in the fixed income markets.

With regulatory constraints resulting in the inability of banks to provide arbitrage capital to the repo market, the Fed took over this function by introducing the Standing Repo Facility (SRF), as discussed in Chapter 11. We will refer to this series of events as an example for an intervention spiral when taking the broader macro-economic perspective of Chapter 20.

While this episode was short-lived, other episodes have been much more significant, lasting for multiple years. For example, throughout the financial crisis of 2008–2009 and the subsequent European sovereign debt crisis and European banking crisis, cross-currency basis swaps between EUR and USD widened from essentially zero to below −40 bp. In other words, European banks that wanted to borrow US dollars with EUR as collateral had to pay an additional 40 bp per year in order to get their counterparties to lend dollars. Contrary to some of the simplified headlines that appeared during this time, there was no "shortage of dollars" in the global markets. Rather, US banks were reluctant to increase the EUR exposure on their balance sheets. And unlike plain interest rate swaps, which are considered "off-balance sheet," cross-currency basis swaps involve full exchange of principal and therefore cause a one-for-one increase in the balance sheets of the banks that conduct them.

In this example, the constraints on bank balance sheets were largely imposed by the banks themselves. But the principle is the same. Balance sheet constraints, self-imposed or imposed by regulators, affect the pricing of instruments in which banks are significant participants. As relative value analysts, we have no choice but to consider these constraints and the effects that they may have on the pricing of the trades we're analyzing.

Short-Term Fluctuations in Supply and/or Demand of Bonds

Given structural changes to various markets in recent years, even large markets, such as that for German Bunds, can be affected by short-term changes in supply or demand. Consider the following story from Reuters, dated February 22, 2022:

> Germany's finance agency has stepped in to ease a bond shortage that developed in the overnight lending market, a market source said, in a sign of stress following the European Central Bank's hawkish pivot and more recently the Ukraine–Russia crisis.
>
> ECB President Christine Lagarde's Feb. 3 refusal to rule out an interest rate hike in 2022 sent traders rushing to repo markets to borrow German bonds to 'short' – essentially to bet prices would fall further as rate rises approach.

> That increased the scarcity of German bonds, the euro zone's safe assets used widely as collateral against repo loans, leading to a plunge in repo rates ...[7]

These market situations may cause richness in some bonds, which cannot be explained by the models, such as the Bunds in Figures 17.1 and 17.2: while some of their richness could be justified by the relatively low CDS of Germany and the haircut schedules, the excessive richness in some sectors may well reflect a specific supply/demand situation. If these imbalances are expected to be short-lived, one could consider positioning for a relative cheapening of Bunds, with their "fair" richness (as given by the models) as a possible target for the trade.

Implications

The pricing of bonds and swaps discussed in previous chapters focused largely on variables that everyone can observe: bond yields, swap rates, repo rates, interbank rates, basis swap spreads, and credit default swap spreads. But the issues discussed in this chapter tend to be quantities that most people aren't able to observe directly, such as the shadow prices of various constraints that some banks may find binding.

Consider the repo spike of September 2019 discussed above. Was there any way for market participants to anticipate that one or more of the constraints faced by JP Morgan were, in fact, binding at the time? Even then, was there any way for market participants to estimate the shadow costs of those constraints, which would allow them to forecast JP Morgan's appetite to lend cash against collateral at various repo rates?

Of course, JP Morgan is only one bank. Presumably, there were other large banks that faced constraints on their balance sheets that affected their willingness to lend. And the typical market participant is just as ignorant of the constraints at other banks as he is of the constraints at JP Morgan.

However, every market participant is able to observe the sum of the impact of all of the complex and individually different regulatory and capital constraints on the market prices in the form of deviations from arbitrage relationships formulated under the assumption of no constraints. Taking the arbitrage inequality formulated in Chapter 16 as an example, the increasing regulatory burdens on banks were visible in the threshold for the arbitrage illustrated in Figure 16.4 increasing from close to the transaction costs before the financial crisis to several dozens of bp, reflecting the particularly high capital requirements for CCBS and CDS. One could hence consider these deviations from

[7] Reuters, "German debt office acts to ease bond shortage after ECB, Ukraine crisis."

arbitrage relationships formulated under the assumption of no constraints as empirically observable variables depending on the empirically unobservable multitude of individually different shadow costs.

So while application of the principles discussed in earlier chapters can help analysts identify instruments that appear rich or cheap to one another, analysts need to at least reflect on the issues discussed in this chapter when predicting whether particular instruments are likely to richen or cheapen in the future. Italian bonds may appear cheap to French bonds on a historical basis, but they may cheapen further if the ECB increases the haircut for Italian bonds still further. German bonds may appear rich to Spanish bonds on a historical basis, but they may richen further in the event that a comment by the ECB President causes a significant number of market participants to borrow bonds via the repo market so they can be sold short in advance of an anticipated ECB hike.

At the same time, the issues discussed in this chapter may be relevant to relative value analysts and traders because of the way their *own* institutions treat these trades. For example, if I work for a financial institution that doesn't face balance sheet constraints, I may be able to take advantage of a relative value opportunity that is priced to reflect the balance sheet constraints of larger banks. For example, if a repo spike causes the basis between Treasury bonds and Treasury bond futures to widen, I may be able to take advantage of the widening by selling bonds against futures and either holding the position until futures' expiration or by unwinding the trade when the repo spike has subsided. One man's constraint may well be another man's opportunity.

CHAPTER 19

Options

INTRODUCTION

A fruitful application of the no-arbitrage principle mentioned in Chapter 1 is the approach of Black and Scholes to option pricing. As the payoff of an option can be replicated by a dynamic self-financing portfolio, the price of an option can be determined by the cost of that portfolio. We start this chapter with a brief sketch of that idea and its implications for option pricing.

However, for trading purposes, delta hedging is particularly important when the gamma of the option is large. This leads to a segmentation of the volatility surface into a sector with high gamma (short expiry), where frequent hedge rebalancing is an important trading strategy, and into a sector with low gamma (long expiry) for which it is not. After introducing this segmentation, we shall discuss the appropriate analysis tools and relative value trades for each of the sectors separately.

A BRIEF REVIEW OF OPTION PRICING THEORY

Since a complete description of option pricing theory and models is outside of the scope of our trade-oriented book, we assume some familiarity of the reader with the basic ideas such as the payoff profile of a call and put. In the following, we briefly highlight just those concepts of option pricing theory, which are of vital importance for finding and exploiting value in option markets.

Delta

The key for both pricing and hedging options is the relationship of the price of an option to the price of its *underlying* (the security the holder of a long option has the right to buy or sell). (In the following examples, these securities will often be swaps on a yield curve.) That relationship is expressed by the number delta, defined as $\frac{\partial PriceOption}{\partial PriceUnderlying}$. As a first derivative of the option price with respect to the value of the underlying, delta tells us the number of units by

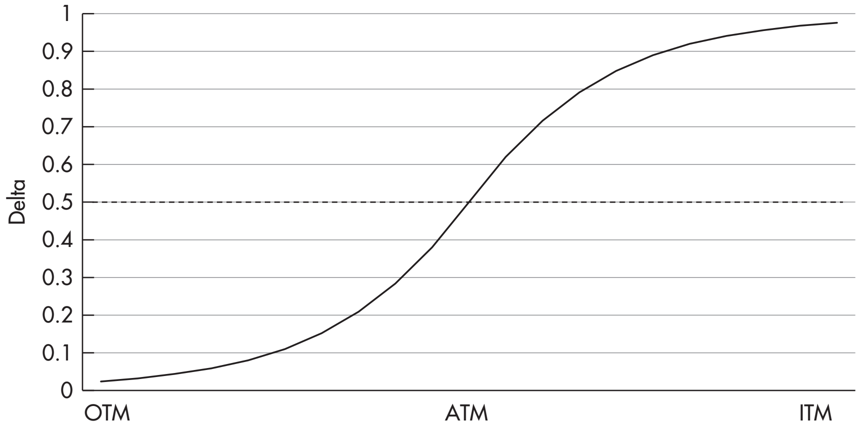

FIGURE 19.1 Delta of an option as a function of the difference between the price of the underlying and the strike price.
Source: Authors.

which the option price would change for a given instantaneous change in the value of the underlying, holding the values of all other inputs constant.

The delta depends in part on the difference between the price of the underlying and the strike price of the option. Figure 19.1 shows schematically the way delta changes with the difference between the price of the underlying and the strike price. For far out-of-the-money (OTM) options, it is close to zero; for far in-the-money (ITM) options, it is close to one; for at-the-money (ATM) options, it is 0.5.[1] Also note that the change in delta (called gamma) is highest for ATM options.

This means that for far OTM options with a delta close to zero, the option price varies only slightly as a result of changes in the price of the underlying. Intuitively, this makes sense, as an option with very little chance of ending ITM will quote close to zero and still quote close to zero even if the underlying changes a bit. Imagine the S&P500 index quotes at 1400 and we have a three-month (3M) option to buy it at 5000. Even if the S&P rises to 1500, our option will still be virtually worthless, hence the change of 100 in the underlying will have produced a negligible change in the option price, which translates into a delta of (almost) zero. On the other side, imagine we have a 3M option to buy the S&P index at 100. At the current price of 1400, this option will be worth about 1300 (plus a tiny premium). If the S&P rises to 1500, the

[1]More precisely, for Black–Scholes $N(d_2) = 0.5$ while $N(d_1)$ could be slightly different.

option value will increase to about 1400. Thus, the change of 100 in the underlying induces a change of about 100 in the option price (i.e. the delta is almost one). Again, this makes sense intuitively, since far ITM options are basically the same as a position in the underlying security.

Delta Hedging

Moreover, delta gives the hedge ratio of an option versus the underlying security. In order to hedge one option, one needs to buy or sell delta of the underlying. For example, to hedge a long ATM call on the S&P500 index we need to sell 0.5 (delta of an ATM option) S&P indices. If the S&P500 index then increases from 1400 to 1500, the option will increase by 100 * delta (0.5) = 50, which is the same amount we lose from being short 0.5 of the underlying.

However, the strike of the option did not change and what was an ATM call at an S&P500 index of 1400 is now an ITM call at an index value of 1500. Consequently, as shown in Figure 19.2, the delta of the option has increased, perhaps to 0.8. This also means that the hedge ratio required has changed. We now need to be short 0.8 of the underlying (i.e. we need to sell an additional 0.3 units of the S&P index). Actually, since delta increased as soon as the S&P500 began to increase from 1400, we were underhedged during the entire move from 1400 to 1500 (with the amount of the underhedge becoming increasingly large as the S&P increased). As a result, we profited from being underhedged in a bull market.

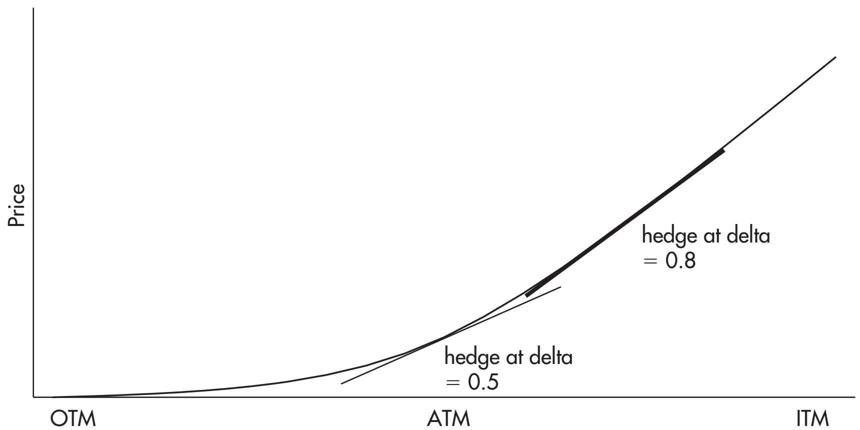

FIGURE 19.2 Price of an option as a function of the difference between the price of the underlying and the strike price.
Source: Authors.